

# The Upper Ahlfors Bound Alone Suffices for the Rough-Operator Pointwise Estimate

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## Source problem and answer

Chamorro, Marcoci, and Marcoci ask whether their estimate

$$T_K^* f(x) \leq C \mathcal{R}_{1,\mu} g(x)$$

for rough kernels can be proved assuming only the upper Ahlfors bound  $\mu(B(x, r)) \leq c_2 r^\nu$ . Their proof assumes two-sided Ahlfors regularity and the extra restriction  $2^{1-\nu} c_2 / c_1 < 1$ . The answer is *yes*: neither the lower Ahlfors bound nor the ratio restriction is needed. The proof below retains the ball-measure factor through the Poincaré–Sobolev step, after which the scale sum is geometric because  $\nu > 1$ .

and in this case the null condition (1.7) is a straightforward consequence of the fact that the function  $\Omega : \mathbb{S}^{n-1} \rightarrow \mathbb{R}$  is such that  $\int_{\mathbb{S}^{n-1}} \Omega \, d\sigma = 0$ . Note now that the condition (1.11) is essentially technical and imposes a special behavior of the measure  $\mu$  and we do not know if it is possible to get rid of this condition. To finish the remarks, we point out that it is an interesting open problem to generalize the estimate (1.12) to a metric measure setting where we *only* dispose the upper Ahlfors condition  $\mu(B(x, r)) \leq c_2 r^\nu$ . This research program will probably require a different approach which it is outside the scope of this article.

## The theorem

Let  $(X, d, \mu)$  have  $0 < \mu(B) < \infty$  for every ball and null metric spheres, as in the source. Write

$$\mathcal{R}_{1,\mu} g(x) = \int_X \frac{d(x, y)}{\mu(B(x, d(x, y)))} g(y) \, d\mu(y).$$

Assume that for some  $q \geq 1$ ,  $C_P > 0$ , and  $\sigma \geq 1$ , every admissible  $f$  and upper gradient  $g$  satisfy

$$\left( \int_{B(x, r)} |f - f_{B(x, r)}|^q \, d\mu \right)^{1/q} \leq C_P r \int_{B(x, \sigma r)} g \, d\mu. \quad (\text{PS})$$

For  $q = 1$ , this has its literal usual meaning.

**Theorem 1** (Upper-Ahlfors resolution). *Suppose  $\nu > 1$  and*

$$\mu(B(x, r)) \leq c_2 r^\nu \quad (x \in X, r > 0). \quad (\text{UA})$$

*Let  $K : X \times X \setminus \{x = y\} \rightarrow \mathbb{R}$  satisfy*

$$|K(x, y)| \leq C_K d(x, y)^{-\nu}, \quad (\text{K1})$$

$$\int_{\{a < d(x, y) < b\}} K(x, y) d\mu(y) = 0 \quad (0 < a < b < \infty). \quad (\text{K2})$$

*Then, wherever the source truncations are defined,*

$$T_K^* f(x) := \sup_{\varepsilon > 0} \left| \int_{\{\varepsilon < d(x, y)\}} K(x, y) f(y) d\mu(y) \right| \leq C \mathcal{R}_{1, \mu} g(x),$$

*where  $C$  depends only on  $C_K, C_P, c_2, \nu, \sigma$ . In particular, the source estimate (1.12) holds under the upper Ahlfors condition alone.*

*Proof.* Fix  $x$ , an inner truncation  $\varepsilon > 0$ , and first impose a finite outer truncation. Choose  $k_0$  with  $2^{k_0-1} < \varepsilon \leq 2^{k_0}$ , and decompose into the first partial annulus  $\{\varepsilon < d(x, y) \leq 2^{k_0}\}$  and the dyadic annuli

$$A_k = \{2^{k-1} < d(x, y) \leq 2^k\}, \quad k > k_0.$$

Put  $B_k = B(x, 2^k)$  and  $B_k^\sigma = B(x, \sigma 2^k)$ . Condition (K2) allows the subtraction of the separate constant  $f_{B_k}$  on every annulus, including the first partial annulus. Therefore (with harmless changes in constants)

$$|T_K^\varepsilon f(x)| \leq C \sum_{k \geq k_0} 2^{-k\nu} \int_{A_k} |f(y) - f_{B_k}| d\mu(y). \quad (1)$$

The displayed estimate can be proved first for finitely many annuli; all subsequent bounds are independent of the outer cutoff.

Set  $\alpha = 1 - 1/q$  when  $q > 1$ . Holder's inequality, the inclusions  $A_k \subset B_k \subset B_k^\sigma$ , and (PS) give

$$\begin{aligned} \int_{A_k} |f - f_{B_k}| d\mu &\leq \mu(B_k)^\alpha \left( \int_{B_k} |f - f_{B_k}|^q d\mu \right)^{1/q} \\ &\leq C_P 2^k \frac{\mu(B_k)}{\mu(B_k^\sigma)} \int_{B_k^\sigma} g d\mu \\ &\leq C_P 2^k \int_{B_k^\sigma} g d\mu. \end{aligned} \quad (2)$$

For  $q = 1$ , (2) follows directly from (PS). This is the decisive point: the two powers of  $\mu(B_k)$  add to one, and the resulting volume ratio is at most one. Substitution into (1) yields

$$|T_K^\varepsilon f(x)| \leq C \sum_{k \in \mathbb{Z}} 2^{k(1-\nu)} \int_{B(x, \sigma 2^k)} g(y) d\mu(y). \quad (3)$$

Extending the nonnegative sum to all integers only enlarges it.

Tonelli's theorem and  $t = d(x, y)$  transform the right side of (3) into

$$C \int_X g(y) \sum_{\{k: \sigma 2^k \geq t\}} 2^{k(1-\nu)} d\mu(y).$$

Since  $\nu > 1$ , the inner series is geometric and

$$\sum_{\{k: \sigma 2^k \geq t\}} 2^{k(1-\nu)} \leq C_{\nu, \sigma} t^{1-\nu}. \quad (4)$$

Finally, (UA) gives

$$t^{1-\nu} \leq c_2 \frac{t}{\mu(B(x, t))}.$$

Combining this with (3)–(4) proves the desired bound, uniformly in both truncations. Passing to the source truncation and taking the supremum over  $\varepsilon$  completes the proof.  $\square$

## Why the source needed a lower bound, and why this proof does not

The source first replaces each annular integral by a normalized whole-ball term. It then has to control  $\sum_k 2^k \mu(B(x, \sigma 2^k))^{-1} \int_{B(x, \sigma 2^k)} g$ , whose inner ball contribution is absorbed using lower Ahlfors growth and condition (1.11); this is the step on pp. 9–10 of the source reproduced in the packet’s audit files.

Estimate (2) avoids that larger series. Its coefficient is instead  $2^{k(1-\nu)}$ , already summable at large scales. Thus the proof also shows that the source’s numerical restriction is not merely improvable by choosing a larger geometric base: it is completely unnecessary.

The hypothesis  $\nu > 1$  is exactly what makes (4) converge and is already implicit in the source’s use of  $\mathcal{R}_{1, \mu}$ , defined for  $0 < 1 < \nu$ . No relation such as  $q < \nu$  is used here. This packet does not alter the source’s separate Morrey estimate for  $\mathcal{R}_{s, \mu}$ , whose proof may still require lower volume information.

## Verification and literature boundary

The proof was checked annulus by annulus, with a finite outer cutoff before Tonelli. Its only volume comparison before the final line is  $\mu(B_k)/\mu(B_k^\sigma) \leq 1$ ; the final conversion uses (UA) in the correct direction. Exact-id and exact-pharse searches through August 11, 2026 found the source and no later paper resolving its stated open problem.

## References

- [1] D. Chamorro, A.-N. Marcoci, and L.-G. Marcoci, *Pointwise estimates for rough operators in a metric measure framework under some Ahlfors regularity conditions*, arXiv:2603.08186 (2026).